

MODULE - 4

JUnit_Basic Testing Exercises

Exercise 1: Setting Up Junit

The screenshot displays the IntelliJ IDEA IDE interface. On the left, the Project Explorer shows a project named 'untitled [JUnitExercise1]' with a directory structure including 'src/main' and 'src/test/java/org.example'. The 'CalculatorTest' class is highlighted in the test directory. The main editor window shows the code for 'CalculatorTest.java', which includes package declarations, imports for JUnit, and a single test method 'testAddition()' that verifies the sum of 2 and 3. The Run toolbar at the bottom indicates that 'CalculatorTest' is being executed. The Run Results panel shows a successful test run with the message '1 test passed 1 test total, 7 ms'. The Run Output panel shows the command used to execute the test: 'C:\Program Files\Java\jdk-25\bin\java.exe' ... and the final status: 'Process finished with exit code 0'.

```
1 package org.example;
2
3 import org.junit.Test;
4 import static org.junit.Assert.*;
5
6 public class CalculatorTest {
7
8     @Test
9     public void testAddition() {
10         int result = 2 + 3;
11         assertEquals("expected: 5, result);", result);
12     }
13 }
```

Run CalculatorTest x

✓ CalculatorTest (org.example) 7 ms ✓ 1 test passed 1 test total, 7 ms

✓ testAddition 7 ms

"C:\Program Files\Java\jdk-25\bin\java.exe" ...

Process finished with exit code 0

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Exercise 2: Basic JUnit Test (CalculatorTest)

The screenshot shows an IDE interface with a project named 'untitled [JUnitExercise1]'. The project structure includes a 'test' directory with a 'java' subdirectory containing 'org.example'. The 'CalculatorTest' class is selected in the project view.

The code editor displays the following Java code for 'CalculatorTest.java':

```
1 package org.example;
2
3 import org.junit.Test;
4 import static org.junit.Assert.*;
5
6 public class CalculatorTest {
7
8     @Test
9     public void testAddition() {
10         int result = 2 + 3;
11         assertEquals("expected: 5, result;", result);
12     }
13 }
```

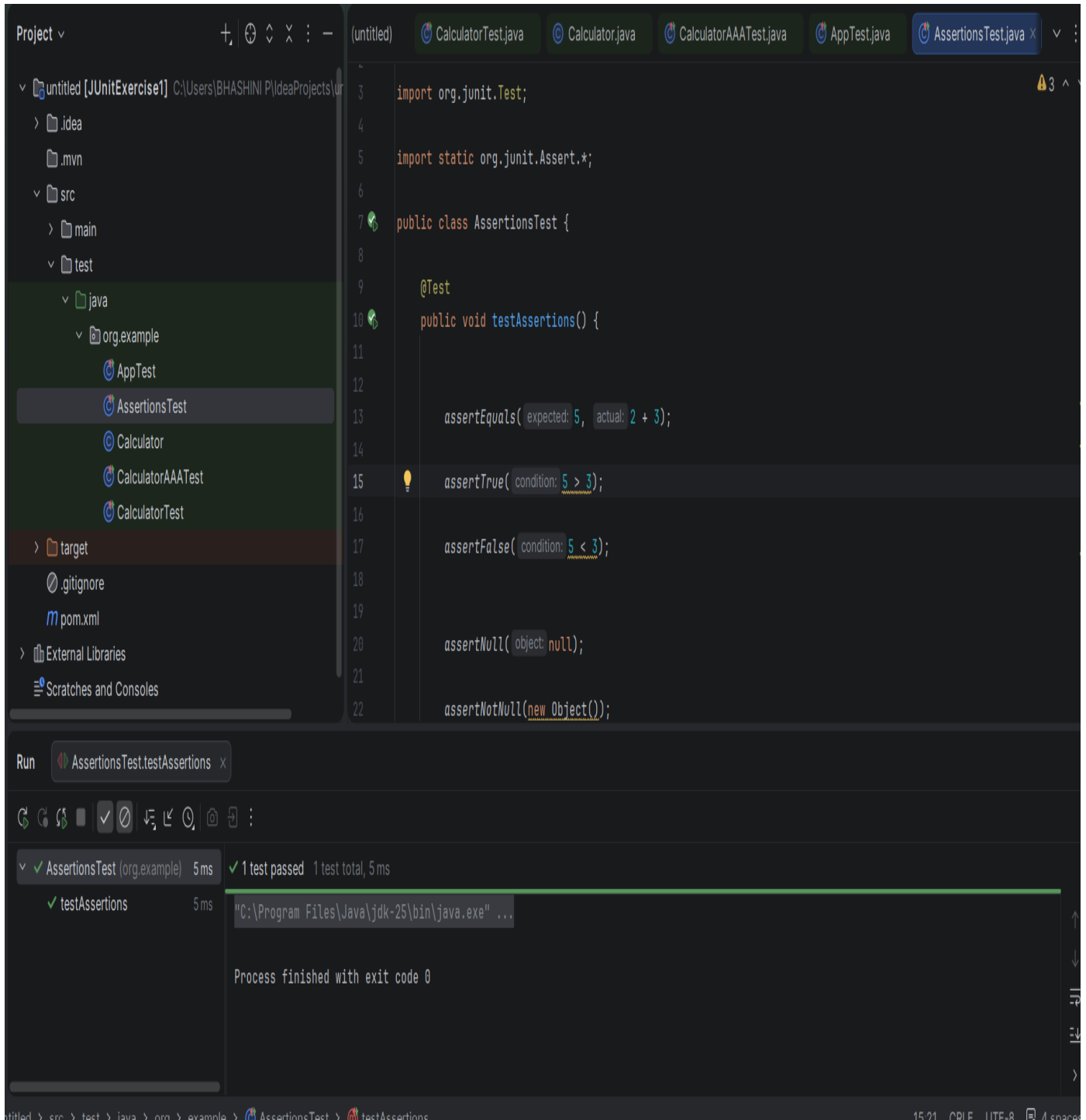
The Run toolbar shows the 'CalculatorTest' test is being executed. The Run output pane displays the following results:

- ✓ CalculatorTest (org.example) 10 ms
- ✓ 1 test passed 1 test total, 10 ms
- ✓ testAddition 10 ms

The console output shows the command: "C:\Program Files\Java\jdk-25\bin\java.exe" ... and the message: "Process finished with exit code 0".

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Exercise 3: Assertions Test



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Exercise 4: @Before and @After (AAA Test)

The screenshot displays the IntelliJ IDEA IDE with a project named "untitled [JUnitExercise1]". The left sidebar shows the project structure, including the "test" directory and the "org.example" package. The "CalculatorAAATest.java" file is selected in the package view.

The main editor shows the code for "CalculatorAAATest.java". The code implements the AAA (Arrange-Act-Assert) test pattern using JUnit annotations:

```
1 package org.example;
2
3 import org.junit.After;
4 import org.junit.Before;
5 import org.junit.Test;
6
7 import static org.junit.Assert.assertEquals;
8
9 public class CalculatorAAATest {
10
11     private Calculator calculator; 3 usages
12
13     @Before
14     public void setUp() {
15         System.out.println("Setting up...");
16         calculator = new Calculator();
17     }
18
19     @After
20     public void tearDown() {
21         System.out.println("Cleaning up...");
22         calculator = null;
23     }
24
25     @Test
26     public void testAdd() {
27
28         int a = 10;
29         int b = 20;
30
31
32         int result = calculator.add(a, b);
33
34         assertEquals("expected: 30, result);
35     }
36 }
37
```

The bottom of the IDE shows the "Run" tab with the command "VerifyInteractionTest.testVerifyInteraction". The output console displays the following results:

```
✓ VerifyInteractionTest 1 sec 934 ms ✓ 1 test passed 1 test total, 1 sec 934 ms
✓ testVerifyInteract(1 sec 934 ms "C:\Program Files\Java\jdk-25\bin\java.exe" ...
```